IN SEARCH OF SAÑCĀRAS: TRADITION-INFORMED REPEATED MELODIC PATTERN RECOGNITION IN CARNATIC MUSIC

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ABSTRACT

Carnatic Music is a South Indian art and devotional musical practice in which melodic patterns (motifs and phrases), known as *sañcāras*, play a crucial structural and expressive role. We demonstrate how the combination of transposition invariant features learnt by a Complex Autoencoder (CAE) and predominant pitch tracks extracted using a Frequency-Temporal Attention Network (FTA-Net) can be used to annotate and group regions of variable-length, repeated, melodic patterns in audio recordings of multiple Carnatic Music performances. These models are trained on novel, expert-curated datasets of hundreds of Carnatic audio recordings and the extraction process tailored to account for the unique characteristics of *sañcāras* in Carnatic Music. Experimental results show that the proposed method is able to identify 54% of all *sañcāras* annotated by a professional Carnatic vocalist. Code to reproduce and interact with these results is available online. [1]

1. INTRODUCTION

Musical styles around the world are often structured in relatively idiosyncratic ways, which can require bespoke tools for MIR tasks and computational music analysis [1]. This is the case in Carnatic Music, a South Indian art and devotional music tradition, in which several characteristic features of the style necessitate tailored MIR approaches.

Relative to other musical styles, Carnatic Music is heavily ornamented. This ornamentation is not superficial decoration but rather is integral to musical meaning [2]. The ornaments, known as *gamakas*, can greatly alter the sound of the notated *svaras* (notes); for example, some *gamakas* do not rest at all on the theoretical pitch of the notated *svara*, and instead involve oscillations between pitches either side of it [3, 4]. This oscillatory movement is particularly characteristic of the style, and can often subsume individual *svaras* [4]. The surface effect on the melodic line is that it typically has fewer stable pitch regions than many other styles. Such qualities makes it impossible for researchers who are not themselves Carnatic musicians to reliably identify *svaras* from audio recordings, although recent work has attempted to automate this task by creating descriptive transcriptions that assign *svara* names to key points in the melodic flow [5–7].

Another important feature of the style is the structural and expressive significance of motifs and phrases known as *sañcāras*, which can be defined as coherent segments of melodic movement that follow the grammar of the *rāga* (melodic framework) [2, 8]. Some musicians use the term in a narrower sense to refer to phrases that are particularly characteristic of the *rāga*, but here we use the term in its broader sense, referring to any melodic segment that is coherent from a Carnatic performer's perspective. These melodic patterns are the means through which the character of the *rāga* is expressed and form the basis of various improvisatory and compositional formats in the style [2,9]. There exists no definitive lists of all possible *sañcāras* in each *rāga*, rather the body of existing compositions and the living oral tradition of *rāga* performance act as repositories for that knowledge.

In this work we take advantage of two deep learning models, CAE [10] and FTA-Net [11], and the Dunya Carnatic corpus [12] for an improved, variable-length, and tradition-informed melodic pattern discovery in Carnatic Music. We first introduce the collections of data we build on top of, and we present our process relating each decision with the tradition-specific characteristics introduced in Section 1.1. We also include an empirical evaluation against expert annotations, brief musicological discussion of results and make available all code, features, models, results and an interactive application to explore them.

1.1 Characteristics of Carnatic Music

Many of the decisions taken in developing the methodology presented here are informed by certain characteristics of Carnatic Music and *sañcāras*, which to varying extents may be shared with other musical traditions around the world. We list some of those most relevant to this task, assigning each a unique code which will be referenced later.

CHAR1. *Sañcāras* conform in some ways to concepts of phrase more widely held; for example in Western Art Music, phrases are understood as influenced by gestalt

[1] https://github.com/MTG/searching_for_sancaras

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principles wherein segments are separated by features such as silence and long periods of stasis [13]. Also, as in other musical styles, longer phrases/*sañcāras* can be understood as comprising shorter but still meaningful segments [14].

CHAR2. The ideal boundary between these shorter segments within a longer phrase is not always clear-cut, as there might be no point of rest or silence between them. Based on discussions with the expert annotator during this project, it is clear that often a longer phrase may be plausibly segmented in two or three different ways. However, there are rules of *rāga* grammar, and understandings of typical *sañcāras* that restrict the number of plausible segmentation points. This is similar to the segmentation of Western Art Music, where in any given section, several plausible segmentations may be made, but where the options are not unlimited [15].

CHAR3. A given *sañcāra* when repeated in a performance may be immediately preceded or followed by different melodic material. This feature is relatively common in many musical styles.

CHAR4. When a *sañcāra* is repeated, it is often elaborated on, which involves the insertion of additional *svaras* and *gamakas*. So the same basic or underlying *sañcāra*/phrase may appear many times in a composition with different elaborations. The commonly occurs in the style because the main compositional format, the *kriti*, has a theme and variation structure.

CHAR5. There can be tempo variations between instances of the same *sañcāra*. This can range from minor fluctuations due to extemporisation, to playing a *sañcāra* at double or half the speed it was originally performed, which is a particular feature of some musical formats such as the *varṇam*.

CHAR6. Single performances within a concert can typically range from between approximately 6 to 60 minutes in length. In the longer examples, the presentation is made up of different musical formats, often involving a composition (e.g., a *kriti*) as well as a number of more extemporised formats (*ālāpana*, *niraval*, *kalpana svaram*), based on *sañcāras* and the *rāga* grammar. This freedom for performers to combine different musical formats to create a larger whole is a typical feature of the style.

CHAR7. A final characteristic feature of the style is its instrumentation. The most widely found contemporary ensemble consists of a vocalist accompanied by violin, *mridangam* (double ended drum) and *tambura* (plucked lute, which creates the drone). Other ensembles led by instrumentalists can also be found.

1.2 Related Work

The automatic retrieval of melodic patterns is a prominent problem in MIR given its applications for music analysis [16], segmentation [17] and music theory [18]. Despite efforts to settle consensus in melodic pattern discovery [19, 20], said task may be built on top of diverse representations of audio signals, such as mel-frequency cepstral coefficients (MFCC) [21], chroma [22], or predominant pitch time-series [23]. In a Carnatic Music context, most existing work relies on processes that consider pairwise distance metrics, such as dynamic time-warping (DTW), between sub-sequences of pitch transcriptions of the predominant sang melody [23–29]. However, this can often be computationally expensive and does not necessarily account for some of the idiosyncrasies of the tradition. To reduce the complexity, several works notate the patterns for an optimized similarity computation [30].

In a Carnatic context, gathering musically-relevant pattern annotations for entire recordings is expensive and time-consuming, and requires the involvement of experts. The most common approach is to ask experts to judge the relevance of the retrieved patterns and evaluate via standard classification metrics [9, 23].

2. DATA

For this research we use the Dunya Carnatic corpus [12], which includes $\approx$ 500 hours or music, divided into 2,380 audio recordings and organized in 235 concerts. Up to 259 artists and 227 unique *rāga*s are present in the corpus. The Saraga dataset is extracted from such corpus and is one of the largest and more complete, open-access datasets for research on the Carnatic and Hindustani music traditions [31]. For a total of 168 Carnatic recordings in Saraga, close-microphone tracks are available. Since the vocalist is the soloist and lead performer in these recordings, for this research we create a sub-dataset comprised of such tracks (168 in total) and denote it Saraga Carnatic Vocal (SCV). All tracks have a sampling rate of 44.1kHz.

We also rely on the Saraga-Carnatic-Melody-Synth (SCMS) dataset [29], a large dataset of vocal melody ground-truth for Carnatic Music compiled using a Carnatic-specific Analysis/Synthesis method, which is currently the largest open source collection of such data and has been shown to positively impact the melodic analysis for Carnatic Music [29].

3. METHOD

Our goal is to extract and group regions of repeated melodic patterns from audio recordings of Carnatic Music performances, Figure 1 provides an overview of this process. Many design decisions are informed by the characteristics introduced in Section 1.1, where relevant, we reference these characteristics using their corresponding **CHARx** code.

3.1 Features

Two feature sets are extracted from each recording in SCV: (1) An automated transcription of the predominant sung pitch in Hz (Section 3.1.1), from which we derive a mask corresponding to silent/stable regions (Section 3.1.1), and (2) Transformation-invariant melodic features extracted using a Complex Autoencoder (CAE) from audio in CQT representation (Section 3.1.2), which we use for self-similarity computation.

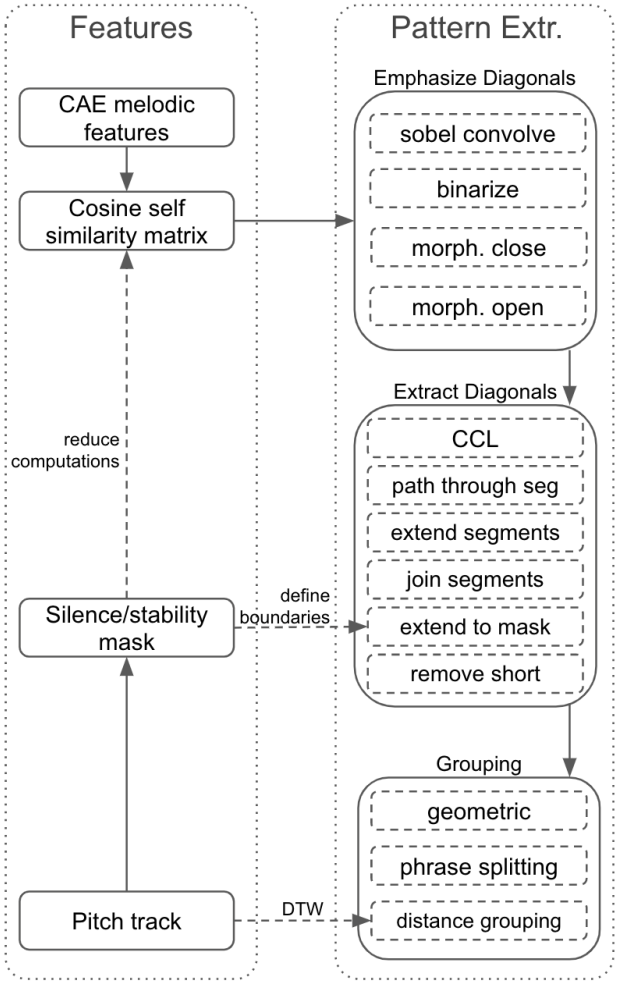


Figure 1. Pipeline overview - two feature sets are extracted from audio (pitch track and CAE features) and used to inform the extraction process.

3.1.1 Pitch Track and Mask

Recent work on vocal pitch extraction for Carnatic Music shows that state-of-the-art heuristic and data-driven methods for such tasks struggle to discriminate between singing voice and violin, and are not able to fully capture the vocal ornamentations in this tradition [29] **[CHAR7]**. To overcome this problem, a frequency-temporal attention network denoted FTA-Net [11] is re-trained using the SCMS dataset, further illustrating the positive impact this has for downstream tasks such as melodic pattern recognition. In this work we take advantage of this procedure to extract cleaner and more versatile pitch tracks from the audio in the Saraga dataset. We prioritize the close-microphone singing voice track for the extraction if available, however, there is no significant performance drop when running the Carnatic-tuned FTA-Net on top of mixed recordings. To account for incorrectly annotated silence that occur within ornaments due, for example, to glottal closure or other rapid vocal movements, we interpolate the pitch tracks to fill gaps shorter than or equal to 250ms [32].

As noted in [29] and [27], sañcāras are unlikely to contain long regions of sustained silence or pitch stability. We identify these regions in our pitch tracks and use this information to inform our search [**CHAR1**].

Silence is annotated as the regions of the pitch track that correspond to 0 Hz, in theory these are all regions where there is no sung voice (remembering that all silences of 250ms or less have been interpolated).

Stability is computed using the method outlined in [27, 29] - for a hop size of 0.2s, a window is passed across the entirety of the pitch track, those windows in which the maximum/minimum frequency is within $\pm$8Hz of it's mean are considered stable. Regions of the pitch track that contain consecutive stable windows whose total length sum to more than 1s are annotated as corresponding to a held note. The remaining windows, stable or not, are left unannotated. The final silence/stability mask, M_{ss}, annotates regions that correspond to silence or a held note with 1, and 0 otherwise.

3.1.2 Melodic Features and Self-similarity

We extract melodic features using a Complex Autoencoder (CAE). Mapping a signal, x, onto complex basis functions learnt by the CAE results in a transformation-invariant "magnitude space", $\mathbf{r}_x$, and a transformation-variant "phase space", ϕ_x. Exploiting the invariance-property of $\mathbf{r}_x$ has proven to achieve state-of-the-art results in repeated section discovery for audio [10].

We train a CAE on the vocal tracks in SCV, represented by their constant-Q transformed spectrogram with a hop size of 1984. The range comprises 120 frequency bins (24 per octave), starting from a minimal frequency of 80 Hz (selected as the minimum frequency expected in Carnatic vocal performance [33]). The spectrogram is split into n-grams of 32 frames. Before training, samples from the training data are randomly transposed by [-24 .. 24] frequency bins or time shifts of [-12 .. 12].

Each audio recording in SCV is mapped onto the complex basis functions learnt by the CAE and the transformation invariant "magnitude space", $\mathbf{r}_x$, used as melodic features relevant to the task of repeated pattern discovery.

As in [10], we compute a self-similarity matrix, $\mathbf{X}$, of $\mathbf{r}_x$, using the reciprocal of the cosine distance (plus some negligible epsilon to avoid division by zero). With some Carnatic performances lasting upwards of one hour, this computation rapidly becomes expensive. We use the masks learnt in Section 3.1.1, M_{ss}, to reduce the computation to include only those areas corresponding to regions we are interested in, reducing the number of pairwise comparisons by around 67%. Typically, $\mathbf{X}$ is between around 10,000 and 50,000 elements squared **[CHAR1, CHAR6]**.

Diagonals in $\mathbf{X}$ of high value correspond to two regions of continued similarity (repeated pattern). It is these diagonals we want identify and extract.

3.2 Repeated Pattern Extraction

Strong diagonal segments in $\mathbf{X}$ correspond to two occurrences of a repeated pattern. We define and extract them using a series of image processing techniques.

3.2.1 *Emphasizing diagonals*

To accentuate the diagonal segments, $\mathbf{X}$ is first convolved with a Sobel Filter to emphasize edges and then binarized about some experimentally set threshold, T_B. Since the emphasized edges correspond to the borders of a diagonal segment and not the segment itself, the binary matrix is morphologically closed to fill gaps and morphologically opened to smooth and remove noisy artifacts. Figure 2 illustrates this process.

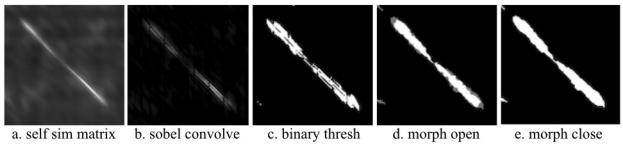


Figure 2. Emphasizing diagonals with convolution and morphological transformations: (a) Self-similarity matrix, (b) convolve with sobel filter, (c) binarize, (d) morphological opening, (e) morphological closing.

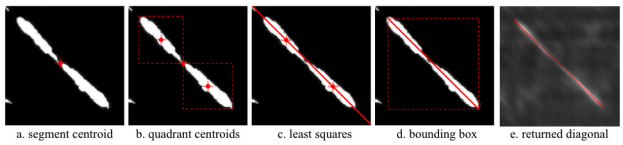


Figure 3. Defining diagonals from non-zero regions. Each segment is a thick region of non-zero values in $\mathbf{X}$, we extract a single line through this region corresponding to the underlying diagonal segment: (a) Identify segment centroid, (b) identify quadrant centroids, (c) least squares line, (d) identify bounding box, (e) return diagonal within bounding box.

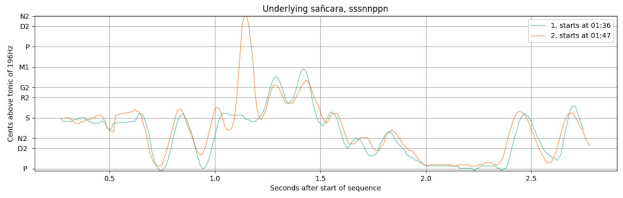


Figure 4. Two instances of the underlying sañcāra, sssnnppn, corresponding to the diagonal segment in Figures 2 and 3

3.2.2 *Extracting Diagonals*

Since instances of the same sañcāra can be performed at differing tempos, segments in $\mathbf{X}$ are not necessarily parallel to the $x = y$ diagonal, as such the method used to extract them in [10, 34] is not appropriate. We instead use a two-pass binary connected-components labelling algorithm (CCL) with a structuring element that considers elements that touch diagonally [35] to identify and group connecting non-zero elements, this can be conceptualized as a *flood fill* algorithm applied to all non-zero regions of $\mathbf{X}$. Each non-zero group corresponds to multiple x, y coordinates, we want to define a single path through this group to nominate as a candidate for the underlying true diagonal segment. First we compute the centroid of these coordinates and split the bounding box into an upper left quadrant and lower right quadrant centered on it. A line is learnt using least squares regression on the points connecting the two quadrant centroids and the underlying diagonal segment defined along that line between the bounding box of the group as a whole (Figure 3) **[CHAR5]**.

Segments are extended along their trajectory for a maximum of up to 50% of their length either side if the similarity in $\mathbf{X}$ for these regions is below some reduced threshold, T_{ext} where $T_{ext} < T_B$ (Figure 5) **[CHAR3]**.

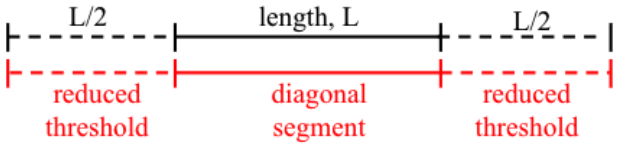


Figure 5. Reducing threshold for significant similarity in regions bordering returned segments. The two regions bordering the trajectory of a returned segment are subject to a lower threshold for similarity to account for variations in preceding and succeeding svaras in sañcāras.

Neighboring segments are joined if they are within 0.5s of each other and their gradients differ by $\leq$0.07rad **[CHAR4]**.

Segments that traverse regions identified in M_{ss} are broken along the center points of these silent/stable regions. Segments that are within 50% of their length to a silent or stable region (annotated in M_{ss}) are extended to the centroid of that silent/stable region **[CHAR1]**.

3.3 Grouping

Relationships between identified patterns are implicit in the geometry of $\mathbf{X}$ - each segment corresponds to two separate instances of the same pattern, those segments that intersect in x correspond to the same region in the recording (that represented on the x-axis) and those in y, vice versa. This information is used to iteratively group all patterns.

3.3.1 *Phrase Splitting*

We can further exploit segment positions in $\mathbf{X}$ to learn boundaries between shorter segments and longer phrases. If two segments intersect on one axis but are of different lengths, new segments are created corresponding to the intersecting portion of the segments **[CHAR2]**. Figure 6 illustrates an example of such. Segment A (corresponding to two patterns, $[x_1 : x_2]$ and $[y_1 : y_2]$) intersects with Segment B (corresponding to two patterns, $[x_1 : x_3]$ and $[y_3 : y_5]$). Consequently, two new segments are created from Segment B, $[y_3 : y_4]$ and $[y_4 : y_5]$ where $[y_3 : y_4]$ is another occurrence of $[x_1 : x_2]$. All patterns corresponding to Segment A, Segment B and the two new sub-segments are included in the final results.

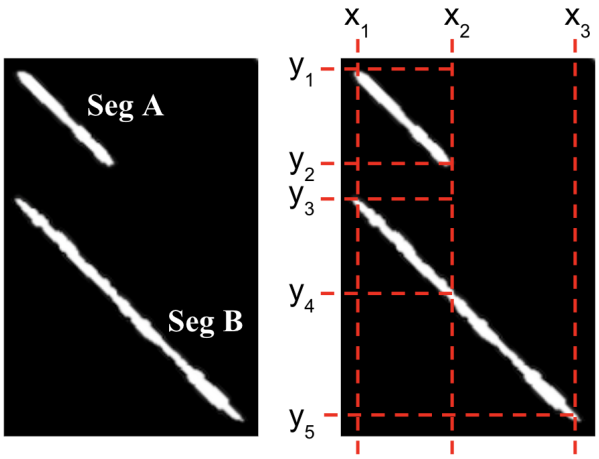


Figure 6. Learning sub-phrase relationships using geometry in $\mathbf{X}$ - Segment B is split into two since a subsequence of Segment B exists in isolation elsewhere (Segment A), both the entire Segment B and it's two subsequences are returned.

Performance	N_r	N_a	Recall	Precision	F1
KJ	174	168	0.73	0.76	0.74
SJ	214	106	0.53	0.54	0.52
VNK	90	144	0.33	0.42	0.37
Overall	478	418	0.54	0.60	0.57

Table 1. Melodic pattern discovery results for the three performances in Table 2. N_r and N_a are the number of returned and number of annotated patterns respectively

3.3.2 Distance Grouping

Further grouping is attempted by comparing each pairwise combination of groups. In each comparison, the dynamic time-warping distance is computed between 10 sets of two randomly selected patterns (one from each group) using the pitch tracks. If the average DTW distance between two groups is below some threshold, T_{dtw}, the groups are considered to represent the same pattern and merged to form one **[CHAR4]**.

4. EXPERIMENTAL SETUP

We apply our approach to three Carnatic performances (Table 2) and compare the results against expert annotations.

4.1 Annotations

Ground-truth annotations of all *sañcāras* in the audio recordings were created by a professional Carnatic vocalist who has 21 years of performance experience in South India. Annotations were created in Carnatic notation, known as *sargam* [36], using the software ELAN [37]. As there is no definitive list of *sañcāras* in a given *rāga*, the segmentations are based on the annotator's experience as a professional performer and student of a highly esteemed musical lineage. These annotations are therefore subjective to some degree, but have the benefit of being based on expert performer knowledge rather than on an externally imposed metric that may be irrelevant to musical concepts held by culture bearers.

In *kritis*, one of the most popular compositional formats, initial phrases are repeated with variations, known as *saṅgatis*. This means that there will be initial *sañcāras* that are subsequently varied. These musical connections are captured by grouping the related musical material together with an 'underlying *sañcāra*' annotation, which refers to the first occurrence and typically simplest version of the *sañcāra*. To reflect the hierarchical nature of plausible musical segmentations, longer full phrase-level annotations that can comprise several shorter *sañcāras* are also created **[CHAR1]**. The evaluation reported here is made on the 'underlying *sañcāra*' and 'underlying full phrase' levels.

4.2 Method and metrics

The various parameters at every step are optimized on one performance, Koti Janmani, representing 35% of the total number of minutes performed across the three performance. Those parameters related to the diagonal segment extraction are selected so as to maximize recall in identifying 'underlying *sañcāra*s' and 'underlying full phrases' that occur more than once in the ground-truth annotations. To consider whether a returned pattern, *R*, is a match with an annotation, *A*, we consider the overlap between them. Since the ideal boundary between shorter segments within a longer phrase is not always clear cut **[CHAR2]**, the expectation of 100% overlap is unrealistic **[CHAR1]**. As in [29], we consider *A* and *R* to be a match if the intersection between them is more than two-thirds the length of *A and* more than two-thirds the length of *R*, inspection of results find this limit to be sufficient in identifying and exploring regions of melodic interest.

5. RESULTS

Table 1 presents the results for the three performances. Our process is able to find 226 of the 418 annotations across the three recordings, corresponding to a total recall of 0.54, precision of 0.60 and F_1 of 0.57. The reader is encouraged to browse and listen to the results using the online visualisation tool provided in our Github repository.

6. DISCUSSION

Although far from 100 percent of annotated patterns were found, we significantly exceed what would have been achieved if we had used out-of-the-box methods without any consideration for the specifities of the Carnatic traditon. For example, when applying the pattern extraction process presented in [10] using our CAE model trained on the SCV, we are able to identify only 8% of annotated patterns at a precision of 7%. It is unsurprising that our performance exceeds this figure since our process is built, in part, on the same underlying model, replacing the pattern extraction steps with more tradition-informed ones.

The results achieved could contribute to a useful tool for musicologists needing to find many instances of the

Alias	Artist	Title	*Rāga*	Composer	Duration
KJ	Akkarai Sisters	Koti Janmani	*Ritigowla*	Oottukkadu Venkata Kavi	08:46
SJ	Salem Gayatri Venkatesan	Sharanu Janakana	*Bilahari*	Purandara Dasa	07:02
VNK	Sumitra Vasudev	Vanajaksha Ninne Kori	*Ritigowla*	Veenai Kuppaiyer	08:37
				Total:	24:25

Table 2. Three Carnatic performances used for evaluation.

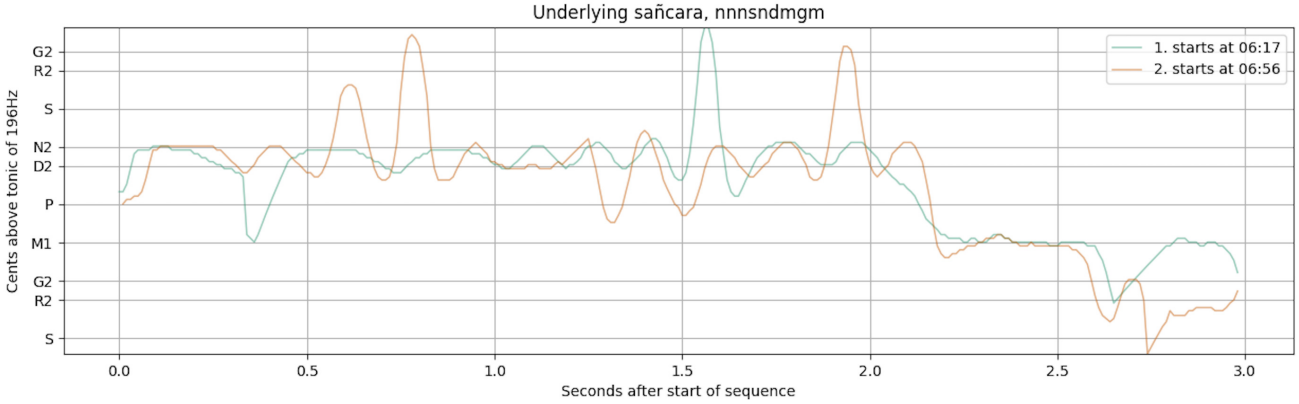


Figure 7. The pattern "nnnsndmgm" in Koti Janmani. The underlying *sañcāra* (blue line) is performed "nnnsndmgm", then later as a distinct variation "nnssndpdnndmmgrgm" (orange line).

same *sañcāra* across a long or many performances, a process that would otherwise involve extremely time consuming labour. While yet to be evaluated formally, the nature of the groupings returned appear promising from a musicological perspective. Figure 4 displays the pitch tracks for the two patterns corresponding to the diagonal segment in Figures 2 and 3. Each pattern would be considered instances of the same underlying sañcāra, "sssnnppn", but in fact are realized slightly differently ("snsgrsnnpn" at 01:36 and "snsmgrsnnpn" at 01:47). The steps outlined in Figures 2 and 3 illustrate how this subtle variation in performance is captured by considering the structural features that are particular to this musical style. However, there comes a point where the variation performed on the underlying or initial *sañcāra* is too great to be found by the process, and indeed it would also not be considered the same *sañcāra* by musicians, but rather a variation or new *saṅgati*. For example, in Koti Janmani, the initial *sañcāra* "nnnsndmgm" is later performed as a distinct variation "nnssndpdnndmmgrgm", and these two are successfully separated by the matching process into two motif groups (11 and 12) (Figure. 7).

While in the *kritis*, Koti Janmani and Sharanu Janakana, the majority of patterns annotated are found, fewer were found in the *varṇam*, Vanajaksha Ninne Kori. This might be explained by the differing structural features of the two compositional styles and of these particular compositions. This *varṇam* has a long 28 beat metrical cycle that lasts approximately 22 seconds. Furthermore, the melodic line has a more discursive and wandering character than the two *kritis* in this analysis. *Kritis* are based on a theme and variation (*saṅgati*) structure, while *varṇams* typically have fewer repetitions of any given phrase. As a result of these qualities, it might be harder for our process to find the borders of segments that match with those of the annotator. As discussed, there is often more than one plausible option for segmentation within longer units. It is possible that even expert annotators would find a larger number of plausible segmentation options in this *varṇam*, something that we could test for in future work by asking two or more annotators to annotate all plausible options, rather than only those that seem optimal **[CHAR2]**.

To demonstrate how our results could be useful to musicologists and fans of the style, we provide a high-level tool to explore and listen to them. We also include the results for 6 more performances across 6 ragas not mentioned here for a total of 9 performances across 8 distinct ragas. We do not have expert annotations for these extra performances and hence cannot empirically evaluate the results. One important factor in how valuable such a tool is, is the extent to which retrieved *sañcāras* are effectively sorted into groups of high melodic similarity that matches musicians' concepts of musical similarity in the style. It is obvious when exploring the results, that even for a listener with little musical background, there is a strong degree of similarity between patterns returned in the same group, both for those patterns that correspond to annotations and otherwise.

7. CONCLUSION

By considering and incorporating the characteristic features of a musical tradition into the development process, we obtain musicologically relevant results for the task of repeated melodic pattern recognition in Carnatic Music. In an effort to provide value for researchers, musicologists and fans of the Carnatic Music tradition, we provide all code, data and a high-level visualisation tool to explore the results.

8. ACKNOWLEDGEMENTS

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